Metrics for the Data Management Campus modules

2026-05-18

This document loads and visualizes data on the usage of modules from the Data Management Campus.
The data are obtained from four sources: the OpenLearnity platform, Matomo, Google Analytics, and Zenodo.
A list of the variables used in the analysis is provided below.

1. Number of logged-in users per week for each module in OpenLearnity

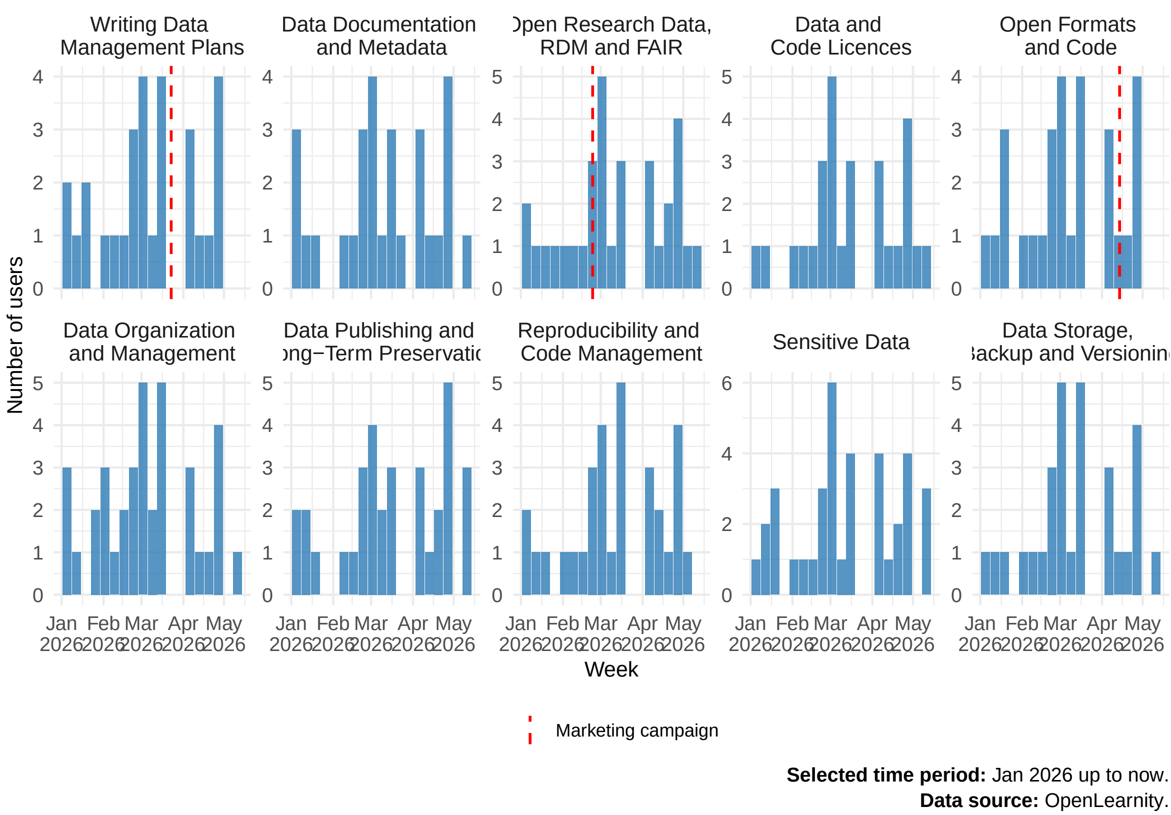
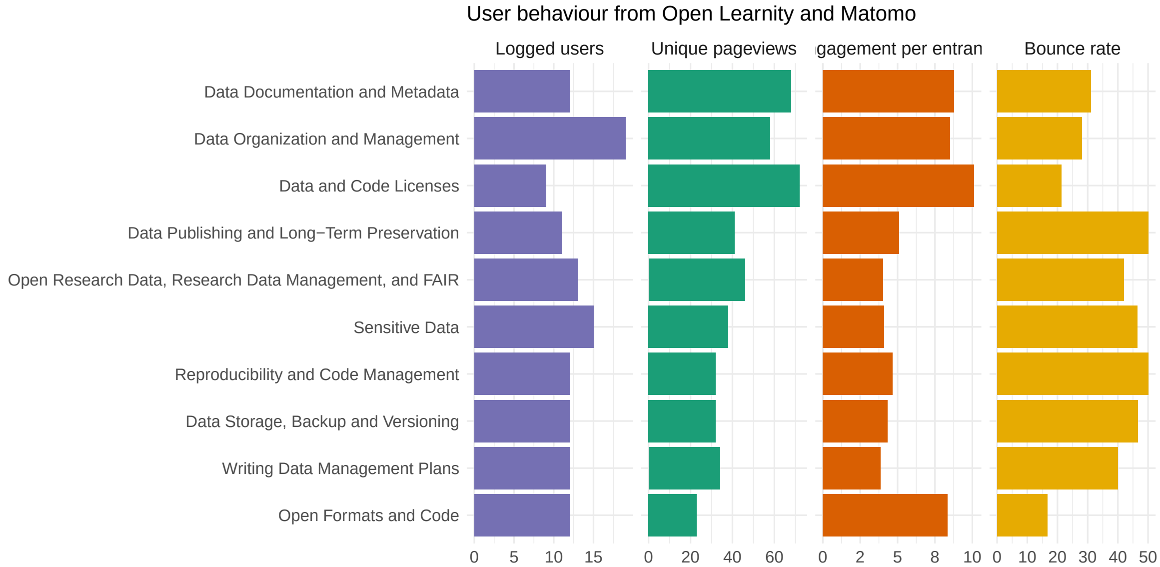


Figure 1 shows weekly user activity across modules based on OpenLearnity enrollment data.
Blue bars represent the *logged_users_week* variable (number of unique users logging in per module each week).
Weekly user activity is generally low, ranging from 0 to about 6 users per week depending on the module, with no strong differences in magnitude across time.
Red dashed lines indicate the timing of the marketing campaign, corresponding to the week when the modules were advertised on ETH Campus Channel screens. No systematic change in activity is visible around the campaign period.

2. User behaviour from Open Learnity and Matomo.



Selected time period: Jan−Mar 2026.
Data source: OpenLearnity and Matomo.

Figure 2 shows user behaviour across modules using Open Learnity user data and Matomo analytics. Four indicators are displayed per module: *logged_users* (Open Learnity enrollment), *unique_pageviews*, *engagement_per_entrance*, and *bounce_rate* (Matomo metrics).

Logged users represent the number of participants enrolled in each module.
Unique pageviews is the number of distint page views of the module's "About" page (counted once per session).
Engagement per entrance is the average number of user interactions after sessions start on the "About" page of a module. This is a proxy for content depth and interest.
Bounce rate is the proportion of single-page sessions without further interaction.

Best-performing module:
The Data and Code Licenses shows the strongest overall engagement in the dataset. It combines a relatively high number of unique pageviews (72) with the highest engagement per entrance (10.1 actions per session), indicating that users interact intensively once they enter the content. In addition, it has one of the lowest bounce rates (21%), suggesting strong retention.

Lowest-performing module:
The Writing Data Management Plans module shows the weakest engagement profile. Although it receives a moderate number of unique pageviews (34), it has the lowest engagement per entrance (3.87 actions per session), indicating limited interaction once users enter the module. Its bounce rate (40%) is also relatively high, suggesting that a significant proportion of users leave without further interactions.

Limitations:
Data is based on the *About* pages and does not capture complete user journeys across the platform, including navigation through home pages or across modules.
The indicators are best interpreted as proxies for interaction intensity rather than direct measures of course progression or learning outcomes.

3. Page views and active users for the ORD pages in Google Analytics.

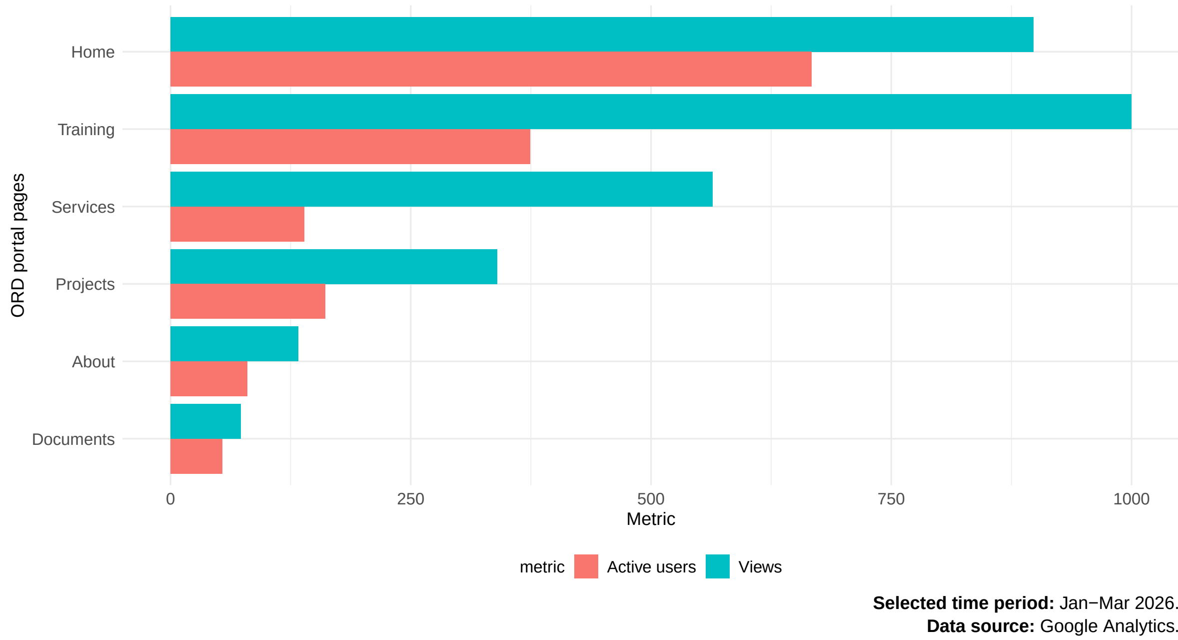


Figure 3 shows the distribution of user engagement across ORD portal pages based on Google Analytics data.
Blue and red bars represent *views* (total number of page views; proxy of intensity of use per page) and *active_users* (number of users whose session lasted at least 10 seconds; proxy of reach per page) for each ORD portal page, obtained from Google Analytics.
The Training page correspond to the Data Management Campus and shows the highest views (1000) but fewer users (374), indicating high engagement per user (strong content usage).

4. Total engagement by resource type and module in Zenodo.

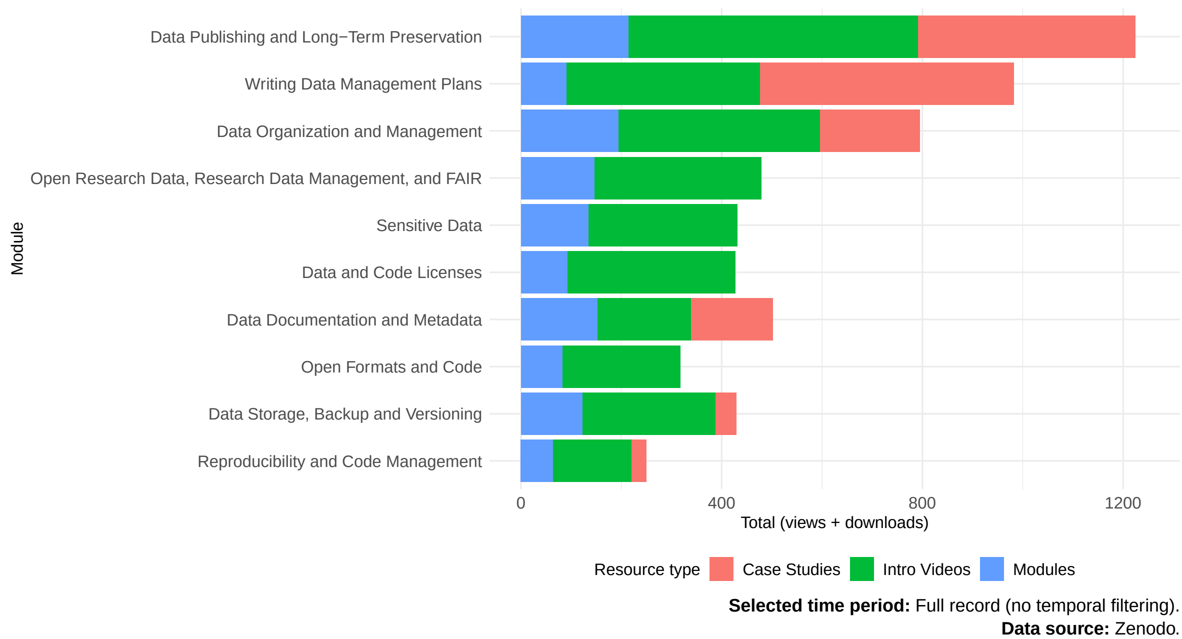


Figure 4 displays the combined total of *views* and *downloads* per resource, as recorded in Zenodo.
Blue, green, and red bars represent different resource types: modules, intro videos, and case studies, respectively. Only 6 of the 10 modules include associated case studies (PowToon videos).
Intro videos account for the majority of total engagement, whereas modules generally receive fewer views and downloads.

5. Number of views and downloads by resource type and module in Zenodo.

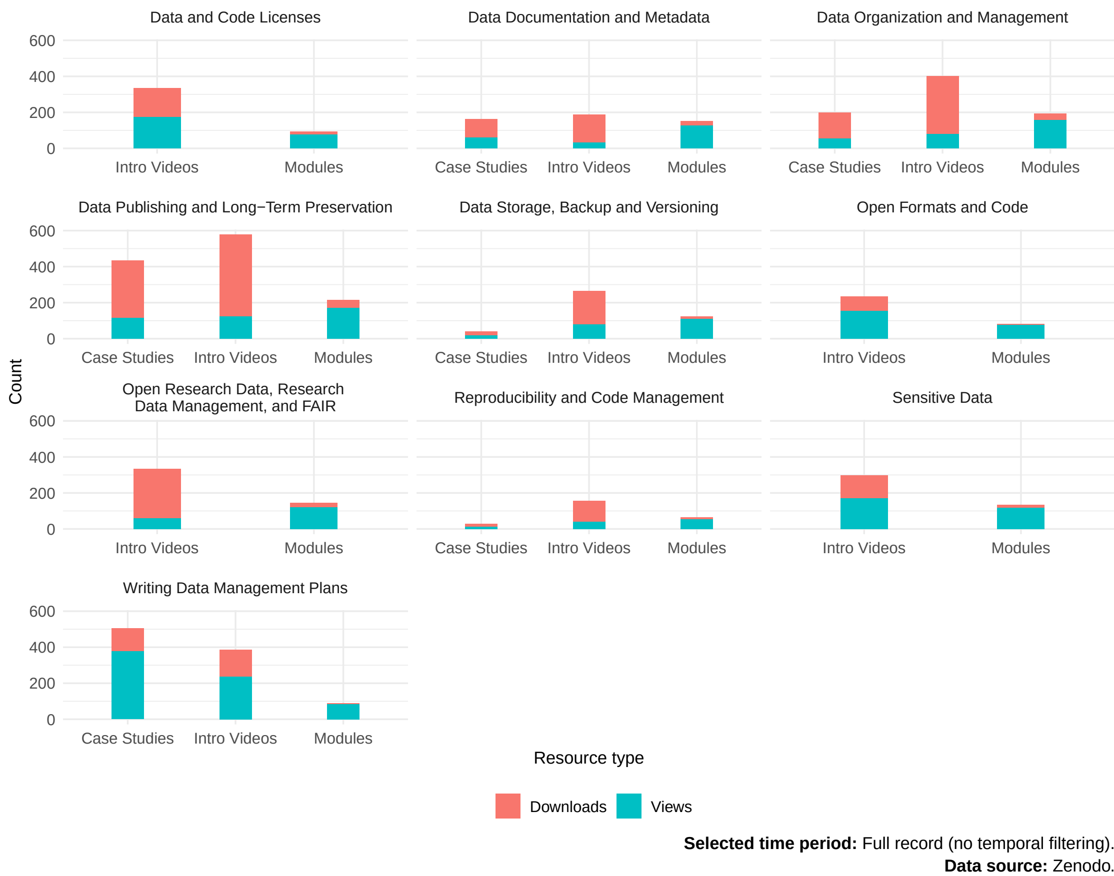


Figure 5 shows the distribution of *views* and *downloads* per resource type (modules, intro videos, and case studies), as recorded in Zenodo.

Blue and red bars represent views and downloads, respectively. Each panel corresponds to a module content, allowing comparison across resource types within the same module. Only 6 of the 10 modules include associated case studies (PowToon videos).

Intro videos account for the majority of total engagement, whereas modules generally receive fewer views and downloads.

6. Number of views and downloads of other resources by institution in Zenodo.

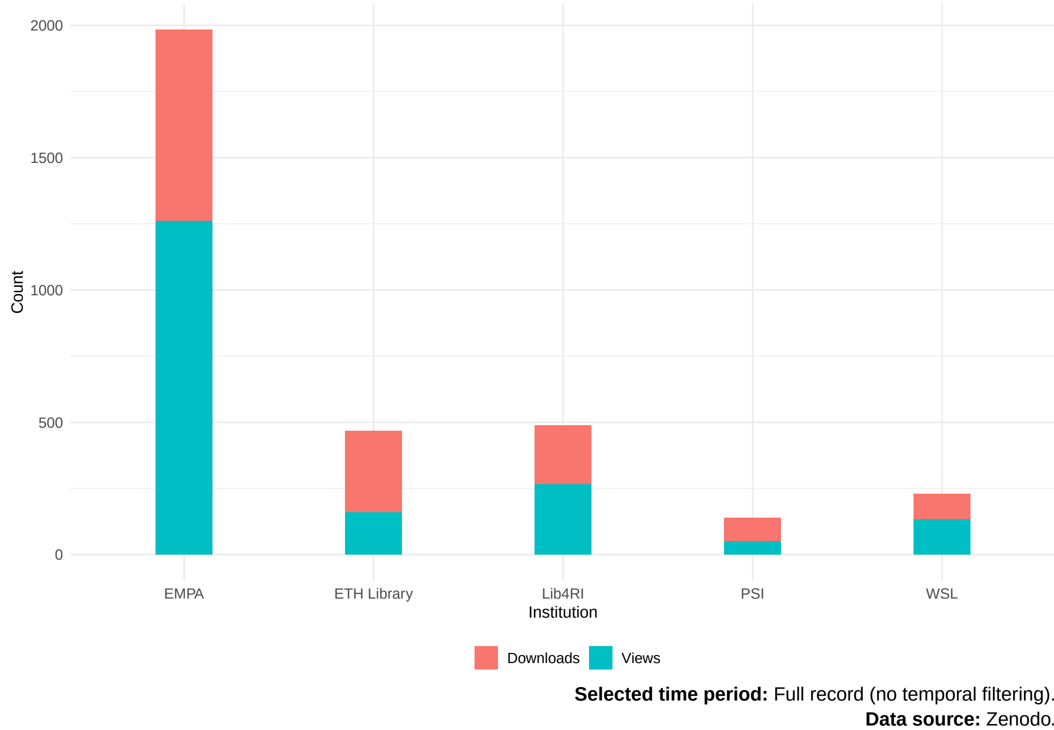


Figure 6 shows the distribution of *views* and *downloads* for other resources (i.e., no module related) in Zenodo, aggregated by institution.
Blue and red bars represent views and downloads, respectively. Values are stacked to show total engagement per institution.
The high views and downloads in EMPA correspond to the Series of video presentation on Research Data Management by Matthias Roesslein.

List of variables used for the analysis of Data Management Campus module usage

indicator	source	estimation	description
user_id	openlearnity	direct measure	user identifier
module_title	openlearnity	direct measure	module title
last_login	openlearnity	direct measure	date of last login in the module. This variable is overwritten each time the same user logs in.
logged_users_week	openlearnity	calculated as count of unique users grouped by week (from last_login) and module	number of unique users per week for each module
logged_users	openlearnity	calculated as count of unique users per module	number of logged users per module during a specific period of time
module_title	matomo_ol	direct measure	module title
unique_pageviews	matomo_ol	direct measure	number of page views counted once per session; unique page views of the module's "About" page
entrances	matomo_ol	direct measure	number of sessions starting on the "About" page
actions_after_entry	matomo_ol	direct measure	number of interactions (clicks, downloads, pageviews) performed after session start on the "About" page
bounces	matomo_ol	direct measure	sessions where the user leaves after viewing only one page
engagement_per_entrance	matomo_ol	calculated as actions_after_entry / entrances	average number of user interactions after session start on the "About" page
bounce_rate	matomo_ol	calculated as bounces / entrances	proportion of visits where users leave after one page
page_title	google_analytics_ord	direct measure	title of the pages in the ORD Portal
views	google_analytics_ord	direct measure	total number of page views; proxy of intensity of use per page
active_users	google_analytics_ord	direct measure	number of users who engaged with the page (session last at least 10 seconds)
record_id	zenodo	direct measure	Zenodo resource identifier
resource_title	zenodo	direct measure	resource title
resource_type	zenodo	direct measure	type of resource (lesson, presentation, video/audio)
views	zenodo	direct measure	number of times resource was viewed
downloads	zenodo	direct measure	number of times resource was downloaded